



8.0.1 UGC NET CSE | June 2013 | Part 3 | Question: 46



Match the following :

- | | |
|---------|---|
| a. TTL | i. High component density |
| b. ECL | ii. Low power consumption |
| c. MOS | iii. Evaluation of "diode-transistor-logic" |
| d. CMOS | iv. High Speed digital circuits |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-iii, b-ii, c-i, d-iv | B. a-i, b-iv, c-iii, d-ii |
| C. a-iii, b-iv, c-i, d-ii | D. a-i, b-ii, c-iii, d-iv |

ugcnetcse-june2013-paper3 digital-logic

Answer key

8.0.2 UGC NET CSE | September 2013 | Part 2 | Question: 26



Match the following identities/laws to their corresponding name :

- | | |
|-------------------------|-----------------|
| (a) $x+x = x$ | i. Dominance |
| $x \cdot x = x$ | |
| (b) $x+0 = x$ | ii. Absorption |
| $x \cdot 1 = x$ | |
| (c) $x+1 = 1$ | iii. Idempotent |
| $x \cdot 1 = x$ | |
| (d) $x \cdot (x+y) = x$ | iv. Identity |

Codes :

- | | |
|-----------------------------------|-----------------------------------|
| A. (a)-iii, (b)-iv, (c)-i, (d)-ii | B. (a)-iv, (b)-iii, (c)-i, (d)-ii |
| C. (a)-iv, (b)-iii, (c)-ii, (d)-i | D. (a)-iii, (b)-iv, (c)-ii, (d)-i |

ugcnetsep2013ii digital-logic match-the-following easy

Answer key

8.0.3 UGC NET CSE | December 2013 | Part 2 | Question: 6



FAN IN of a component A is defined as

- Number of components that can call or pass control to component A.
- Number of components that are called by component A.
- Number of components related to component A.
- Number of components dependent on component A.

ugcnetcse-dec2013-paper2 digital-logic

Answer key

8.0.4 UGC NET CSE | December 2013 | Part 2 | Question: 31



The dual of a Boolean expression is obtained by interchanging

- Boolean sums and Boolean products
- Boolean sums and Boolean products or interchanging $0's$ and $1's$
- Boolean sums and Boolean products and interchanging $0's$ and $1's$
- Interchanging $0's$ and $1's$

Answer key **8.0.5 UGC NET CSE | December 2013 | Part 3 | Question: 51**

Synchronization is achieved by a timing device called a _____ which generates a periodic train of _____.

- A. clock generator, clock pulse
 C. generator, clock
 B. master generator, clock pulse
 D. master clock generator, clock pulse

ugcnetcse-dec2013-paper3 digital-logic

Answer key **8.0.6 UGC NET CSE | July 2016 | Part 2 | Question: 6**

Which of the following logic expression is incorrect ?

- A. $1 \oplus 0 = 1$
 C. $1 \oplus 1 \oplus 0 = 1$
 B. $1 \oplus 1 \oplus 1 = 1$
 D. $1 \oplus 1 = 0$

ugcnetcse-july2016-paper2 digital-logic

Answer key **8.0.7 UGC NET CSE | July 2016 | Part 2 | Question: 8**

Simplified Boolean equation for the following truth table is :

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

- A. $F = yz + \bar{y}z$
 C. $F = \bar{x}z + xz$
 B. $F = xy + \bar{x}y$
 D. $F = \bar{x}z + xz + xyz$

ugcnetcse-july2016-paper2 digital-logic

Answer key **8.0.8 UGC NET CSE | December 2008 | Part 2 | Question: 13**

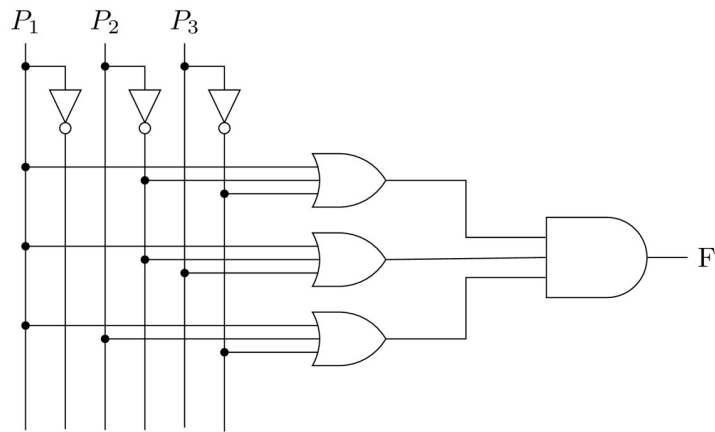
The dual of the switching function $x + yz$ is :

- A. $x + yz$
 C. $x(y + z)$
 B. $x' + y'z'$
 D. $x'(y' + z')$

ugcnetcse-dec2008-paper2 digital-logic dual-function

Answer key **8.0.9 UGC NET CSE | November 2017 | Part 2 | Question: 10**

The output of the following combinational circuit is F .



The value of F is

- A. $P_1 + P_2'P_3$
 B. $P_1 + P_2'P_3'$
 C. $P_1 + P_2P_3'$
 D. $P_1' + P_2P_3$

ugcnetcse-nov2017-paper2

Answer key

8.0.10 UGC NET CSE | December 2018 | Part 2 | Question: 11



Consider the following boolean equations:

- i. $wx + w(x + y) + x(x + y) = x + wy$
 ii. $(w\bar{x}(y + xz) + \bar{w}\bar{x})y = \bar{x}y$

What can you say about the above equations ?

- A. (i) is true and (ii) is false
 B. (i) is false and (ii) is true
 C. Both (i) and (ii) are true
 D. Both (i) and (ii) are false

ugcnetcse-dec2018-paper2 digital-logic

Answer key

8.0.11 UGC NET CSE | December 2018 | Part 2 | Question: 10



In computers, subtraction is generally carried out by

- A. 9's complement
 B. 1's complement
 C. 10's complement
 D. 2's complement

ugcnetcse-dec2018-paper2 digital-logic

Answer key

8.0.12 UGC NET CSE | December 2018 | Part 2 | Question: 5



Which of the following statements are true ?

- i. Every logic network is equivalent to one using just NAND gates or just NOR gates.
 ii. Boolean expressions and logic networks correspond to labelled acyclic digraphs.
 iii. No two Boolean algebras with n atoms are isomorphic.
 iv. Non-zero elements of finite Boolean algebras are not uniquely expressible as joins of atoms

Choose the correct answer from the code given below :

- A. i and iv only
 B. i, ii and iii only
 C. i and ii only
 D. ii, iii, and iv only

ugcnetcse-dec2018-paper2 digital-logic

8.0.13 UGC NET CSE | December 2006 | Part 2 | Question: 9

The characteristic equation of the D flip-flop is :

- A. B. $Q=D$ C. $Q=1$ D. $Q=0$

ugcnetcse-dec2006-paper2

Answer key

8.0.14 UGC NET CSE | December 2006 | Part 2 | Question: 8

A sum of products expression can be implemented with _____ logic gates.

- A. AND-OR B. NAND-OR C. AND-NOT D. OR-AND

ugcnetcse-dec2006-paper2 digital-logic

Answer key

8.0.15 UGC NET CSE | December 2006 | Part 2 | Question: 7

An example of a self complementing code is :

- A. 8421 code B. Gray code C. Excess-3 code D. 7421 code

ugcnetcse-dec2006-paper2

8.0.16 UGC NET CSE | December 2007 | Part 2 | Question: 7

Simplified form of Boolean expression $xy+(\sim x)z+yz$ is :

- A. $xy+(\sim x)z$ B. $(\sim x)y+(\sim x)z$ C. $(\sim x)y+xz$ D. $xy+xz$

ugcnetcse-dec2007-paper2

Answer key

8.0.17 UGC NET CSE | October 2020 | Part 2 | Question: 34

The period of a signal is 100 ms, then the frequency of this signal in kilohertz is _____

- A. 10 B. 10^{-1} C. 10^{-2} D. 10^{-3}

ugcnetcse-oct2020-paper2 digital-logic

Answer key

8.1 Adder (4)**8.1.1 Adder: UGC NET CSE | August 2016 | Part 2 | Question: 10**

Match the following :

List-I

- a. Controlled inverter
b. Full adder
c. Half adder
d. Binary adder

List-II

- i. a circuit that can add 3 bits
ii. a circuit that can add two binary numbers
iii. a circuit that transmits a binary word or its 1's complement
iv. a logic circuit that adds 2 bits

Codes :

- A. a-iii, b-ii, c-iv, d-i
C. a-iii, b-iv, c-i, d-ii
B. a-ii, b-iv, c-i, d-ii
D. a-iii, b-i, c-iv, d-ii

ugcnetcse-aug2016-paper2 digital-logic adder

Answer key

8.1.2 Adder: UGC NET CSE | December 2005 | Part 2 | Question: 7



A half-adder is also known as :

- A. AND Circuit B. NAND Circuit C. NOR Circuit D. EX-OR Circuit

ugcnetcse-dec2005-paper2 combinational-circuit adder

Answer key

8.1.3 Adder: UGC NET CSE | December 2013 | Part 2 | Question: 40



Which of the following statements are true ?

- I. A circuit that adds two bits, producing a sum bit and carry bit is called half adder.
II. A circuit that adds two bits, producing a sum bit and carry bit is called full adder.
III. A circuit that adds two bits and a carry bit producing a sum bit and a carry bit is called full adder.
IV. A device that accepts the value of a Boolean variable as input and produces its complement is called an inverter.
- A. I & II B. II & III C. I, II, III D. I, III & IV

ugcnetcse-dec2013-paper2 digital-logic adder

Answer key

8.1.4 Adder: UGC NET CSE | December 2014 | Part 2 | Question: 06



The BCD adder to add two decimal digits needs minimum of

- A. 6 full adders and 2 half adders B. 5 full adders and 3 half adders
C. 4 full adders and 3 half adders D. 5 full adders and 2 half adders

ugcnetcse-dec2014-paper2 digital-logic adder

Answer key

8.2

Boolean Algebra (10)

8.2.1 Boolean Algebra: UGC NET CSE | December 2009 | Part 2 | Question: 06



The simplified form of the Boolean expression $(X + Y + XY)(X + Z)$ is

- (A) $X + Y + ZX + Y$
(B) $XY - YZ$
(C) $X + YZ$
(D) $XZ + Y$

ugcnetcse-dec2009-paper2 digital-logic boolean-algebra

Answer key

8.2.2 Boolean Algebra: UGC NET CSE | December 2018 | Part 2 | Question: 7



The boolean expression $\bar{A} \cdot B + A \cdot \bar{B} + A \cdot B$ is equivalent to

- A. $\bar{A} \cdot B$ B. $\overline{A+B}$ C. $A \cdot B$ D. $A+B$

ugcnetcse-dec2018-paper2 digital-logic boolean-algebra

Answer key

8.2.3 Boolean Algebra: UGC NET CSE | July 2016 | Part 2 | Question: 9



The simplified form of the Boolean equation $(AB + AB + AC)(\bar{A}C + B)$ is

- A. $A\bar{B}$ B. $A\bar{B}C$ C. $\bar{A}B$ D. ABC

Answer key **8.2.4 Boolean Algebra: UGC NET CSE | June 2005 | Part 2 | Question: 7**

The logic expression $\bar{x}y\bar{z} + \bar{x}yz + xy\bar{z} + xyz$ reduces to :

- A. $\bar{x}z$ B. xyz C. y D. yz

ugcnetcse-june2005-paper2 digital-logic boolean-algebra

Answer key **8.2.5 Boolean Algebra: UGC NET CSE | June 2009 | Part 2 | Question: 03**

The Boolean expression $\bar{x}yz + yz + xz$ is equivalent to :

- A. x B. y C. z D. $x+y+z$

ugcnetcse-june2009-paper2 digital-logic boolean-algebra

Answer key **8.2.6 Boolean Algebra: UGC NET CSE | June 2011 | Part 2 | Question: 4**

The absorption law in Boolean algebra say that

- A. $X + X = X$ B. $X.X = X$
C. $X + X.Y = X$ D. None of the above

digital-logic boolean-algebra ugcnetcse-june2011-paper2

Answer key **8.2.7 Boolean Algebra: UGC NET CSE | June 2013 | Part 2 | Question: 41**

How many different Boolean functions of degree 4 are there?

- A. 2^4 B. 2^8 C. 2^{12} D. 2^{16}

ugcnetcse-june2013-paper2 digital-logic boolean-algebra

Answer key **8.2.8 Boolean Algebra: UGC NET CSE | June 2014 | Part 2 | Question: 16**

How many different truth tables of the compound propositions are there that involve the propositions p & q ?

- A. 2 B. 4 C. 8 D. 16

ugcnetcse-june2014-paper2 digital-logic boolean-algebra

Answer key **8.2.9 Boolean Algebra: UGC NET CSE | June 2014 | Part 2 | Question: 17**

A Boolean function F is called self dual if and only if $F(x_1, x_2, \dots, x_n) = F(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$. How many Boolean functions of degree n are self-dual ?

- A. 2^n B. $(2)^{2^n}$ C. $(2)^{n^2}$ D. $(2)^{2^{n-1}}$

ugcnetcse-june2014-paper2 digital-logic boolean-algebra

Answer key **8.2.10 Boolean Algebra: UGC NET CSE | Junet 2015 | Part 2 | Question: 6**

Consider the following statements:

- i. Boolean expressions and logic gates networks correspond to labelled acyclic digraphs
- ii. Optimal boolean expressions may not correspond to simplest networks.
- iii. Choosing essential blocks first in a Karnaugh map and then greedily choosing the largest remaining blocks to cover may not give an optimal expression

Which of these statement(s) is/are correct?

- A. i only B. ii only C. i and ii D. i, ii, and iii

ugcnetcse-june2015-paper2 digital-logic boolean-algebra

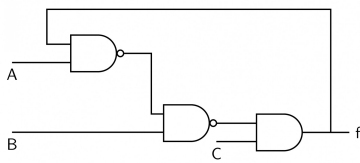
Answer key 

8.3 Circuit Output (5)

8.3.1 Circuit Output: UGC NET CSE | December 2012 | Part 2 | Question: 1



Consider the circuit shown below. IN a certain steady state, 'Y' is at logical '1'. What are the possible values of A, B, C?



- A. $A=0, B=0, C=1$ B. $A=0, B=C=1$ C. $A=1, B=C=0$ D. $A=B=1, C=1$

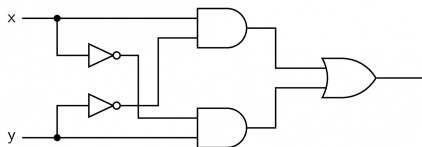
ugcnetcse-dec2012-paper2 digital-logic circuit-output

Answer key 

8.3.2 Circuit Output: UGC NET CSE | December 2013 | Part 3 | Question: 53



What will be output of the following logic diagram ?



- A. $x \text{ OR } y$ B. $x \text{ AND } y$ C. $x \text{ XOR } y$ D. $x \text{ XNOR } y$

ugcnetcse-dec2013-paper3 digital-logic circuit-output

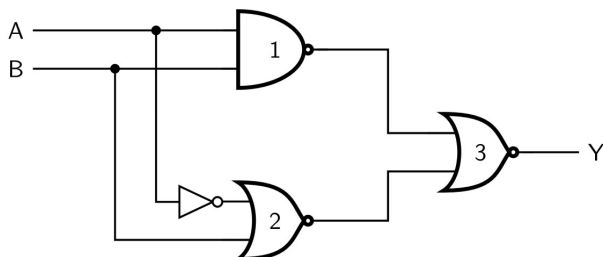
Answer key 

8.3.3 Circuit Output: UGC NET CSE | December 2018 | Part 2 | Question: 17



Find the boolean expression for the logic circuit shown below:

(1-NAND gate, 2-NOR gate, 3-NOR gate)

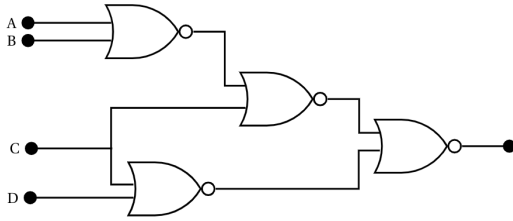


- A. $A\bar{B}$ B. $\bar{A}B$
C. AB D. $\bar{A}\bar{B}$

Answer key

8.3.4 Circuit Output: UGC NET CSE | June 2010 | Part 2 | Question: 6

The logic expression for the output of the circuit shown in the figure is



- A. $\bar{A}\bar{C} + \bar{B}\bar{C} + CD$
 C. $ABC + \bar{C}D$

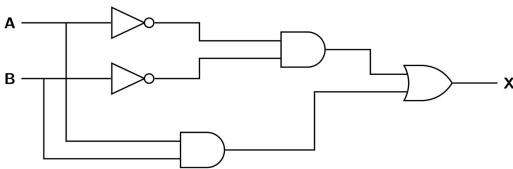
- B. $A\bar{C} + B\bar{C} + \bar{C}D$
 D. $\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{D}$

ugcnetcse-june2010-paper2 digital-logic circuit-output digital-circuits

Answer key

8.3.5 Circuit Output: UGC NET CSE | September 2013 | Part 2 | Question: 49

What type of logic circuit is represented by the figure shown below ?



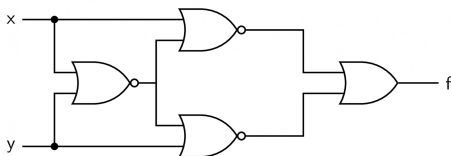
- A. XOR B. XNOR C. XAND D. XNAND

ugcnetsep2013iii digital-logic circuit-output

Answer key

8.4 Combinational Circuit (3)**8.4.1 Combinational Circuit: UGC NET CSE | August 2016 | Part 2 | Question: 9**

Which of the following logic operations is performed by the following given combinational circuit ?



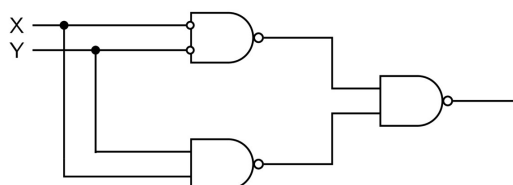
- A. EXCLUSIVE-OR B. EXCLUSIVE-NOR C. NAND D. NOR

ugcnetcse-aug2016-paper2 digital-logic combinational-circuit

Answer key

8.4.2 Combinational Circuit: UGC NET CSE | August 2016 | Part 3 | Question: 6

The output of the following combinational circuit



is :

- A. XY B. $X + Y$ C. $X \oplus Y$ D. $X \oplus Y$

ugcnetcse-aug2016-paper3 digital-logic combinational-circuit

Answer key 

8.4.3 Combinational Circuit: UGC NET CSE | July 2018 | Part 2 | Question: 93



Match the terms in **List-I** with the options given in **List-II** :

- | List-I | List-II |
|--------------------|-------------------------------------|
| (a) Decoder | (i) 1 line to 2^n lines |
| (b) Multiplexer | (ii) n lines to 2^n lines |
| (c) De multiplexer | (iii) 2^n lines to 1 line |
| | (iv) 2^n lines to 2^{n-1} lines |

Code :

- A. (a)-(ii); (b)-(i); (c)-(iii) B. (a)-(ii); (b)-(iii); (c)-(i)
C. (a)-(ii); (b)-(i); (c)-(iv) D. (a)-(iv); (b)-(ii); (c)-(i)

ugcnetcse-july2018-paper2 digital-logic combinational-circuit

Answer key 

8.5 Digital Circuits (10)

8.5.1 Digital Circuits: UGC NET CSE | August 2016 | Part 3 | Question: 65



Which of the following 2 input Boolean logic functions is linearly inseparable ?

- (a) AND
(b) OR
(c) NOR
(d) XOR
(e) NOT XOR

- A. (a) and (b) B. (b) and (c) C. (c), (d) and (e) D. (d) and (e)

ugcnetcse-aug2016-paper3 digital-logic digital-circuits

Answer key 

8.5.2 Digital Circuits: UGC NET CSE | December 2008 | Part 2 | Question: 4



Extremely low power dissipation and low cost per gate can be achieved in:

- A. MOS ICs B. C MOS ICs
C. TTL ICs D. ECL ICs

ugcnetcse-dec2008-paper2 digital-logic digital-circuits

8.5.3 Digital Circuits: UGC NET CSE | December 2008 | Part 2 | Question: 5



An example of a universal building block is:

1. $X - OR$ gate
2. AND gate
3. OR gate
4. NOR gate

ugcnetcse-dec2008-paper2 digital-logic digital-circuits

Answer key 

A. $A \oplus B$
C. $(A \oplus B) \odot A$

- B. $A \odot B$
D. $(A \odot B) \oplus A$

Answer key 

- a. TTL
- b. ECL
- c. CMOS
- i. NAND
- ii. NOR
- iii. Inverter

A. a-i, b-ii, c-iii
C. a-ii, b-iii, c-i

- B. a-iii, b-ii, c-i
D. a-ii, b-i, c-iii

Answer key

a. TTL	i. High fan out
b. ECL	ii. Low propagation delay
d. CMOS	iii. High power dissipation

A. a-iii, b-ii, c-i
C. a-i, b-iii, c-ii

- B. a-i, b-ii, c-iii
D. a-iii, b-i, c-ii

Answer key 

A. OR B. NOR C. EX-OR D. NAND

Answer key 

A. Race condition B. Saturation C. Delay D. High impedance

Answer key 

A. $1V$ B. $2V$ C. $5V$ D. $10V$

8.5.10 Digital Circuits: UGC NET CSE | September 2013 | Part 2 | Question: 23

Which one of the following set of gates is best suited for 'parity' checking and 'parity' generation?

- A. AND, OR, NOT B. NAND, NOR C. EX-OR, EX-NOR D. None of the above

ugcnetsep2013iii digital-logic digital-circuits

Answer key

8.6 Digital Counter (2)**8.6.1 Digital Counter: UGC NET CSE | December 2007 | Part 2 | Question: 8**

In order to build a MOD - 18 counter, the minimum number of flip flops needed is equal to :

- A. 18 B. 9 C. 5 D. 4

ugcnetcse-dec2007-paper2 digital-logic digital-counter

Answer key

8.6.2 Digital Counter: UGC NET CSE | June 2019 | Part 2 | Question: 15

What will be the number of states when a MOD- 2 counter is followed by a MOD- 5 counter?

- A. 5 B. 10 C. 15 D. 20

ugcnetcse-june2019-paper2 digital-logic digital-counter

Answer key

8.7 Encoding (1)**8.7.1 Encoding: UGC NET CSE | June 2014 | Part 2 | Question: 37**

A text is made up of the characters $\alpha, \beta, \gamma, \delta$ and σ with the probability 0.12, 0.40, 0.15, 0.08 and 0.25 respectively. The optimal coding technique will have the average length of

- A. 1.7 B. 2.15 C. 3.4 D. 3.8

ugcnetcse-june2014-paper2 computer-networks encoding

Answer key

8.8 Excess 3 (1)**8.8.1 Excess 3: UGC NET CSE | December 2014 | Part 2 | Question: 07**

The Excess-3 decimal code is a self-complementing code because

- A. The binary sum of a code and its $9's$ complement is equal to 9.
 B. It is a weighted code.
 C. Complement can be generated by inverting each bit pattern.
 D. The binary sum of a code and its $10's$ complement is equal to 9.

ugcnetcse-dec2014-paper2 digital-logic excess-3

Answer key

8.9 Flip Flop (10)

8.9.1 Flip Flop: UGC NET CSE | December 2004 | Part 2 | Question: 10

The characteristic equation of a T flip-flop is :

- A. $Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n$
- B. $Q_{n+1} = T + Q_n$
- C. $Q_{n+1} = TQ_n$
- D. $Q_{n+1} = \bar{T}Q_n$

The symbols used have the usual meaning.

ugcnetcse-dec2004-paper2 flip-flop

Answer key

8.9.2 Flip Flop: UGC NET CSE | December 2005 | Part 2 | Question: 10

When an inverter is placed between both inputs of an S-R flip flop, the resulting flip flop is :

- A. JK flip-flop
- B. D flip-flop
- C. T flip-flop
- D. None of these

ugcnetcse-dec2005-paper2 digital-logic flip-flop

Answer key

8.9.3 Flip Flop: UGC NET CSE | December 2008 | Part 2 | Question: 14

The characteristics equation of D flip-flop is :

- A. $Q=1$
- B. $Q=0$
- C. $Q=\bar{D}$
- D. $Q=D$

ugcnetcse-dec2008-paper2 digital-logic flip-flop

Answer key

8.9.4 Flip Flop: UGC NET CSE | December 2010 | Part 2 | Question: 8

An Astable multivibrator has :

- A. One stable state
- B. Two stable states
- C. No stable states
- D. None of these

ugcnetcse-dec2010-paper2 digital-logic flip-flop

Answer key

8.9.5 Flip Flop: UGC NET CSE | January 2017 | Part 2 | Question: 8

A binary 3-bit down counter uses J-K flip-flops, FF_i with inputs J_i, K_i and outputs $Q_i, i = 0, 1, 2$ respectively. The minimized expression for the input from following is :

- I. $J_0 = K_0 = 0$
- II. $J_0 = K_0 = 1$
- III. $J_1 = K_1 = Q_0$
- IV. $J_1 = K_1 = \bar{Q}_0$
- V. $J_2 = K_2 = Q_1 Q_0$
- VI. $J_2 = K_2 = \bar{Q}_1 Q_0$

- A. I, III, V
- B. I, IV, VI
- C. II, III, V
- D. II, IV, VI

ugcnetjan2017ii digital-logic flip-flop

Answer key

8.9.6 Flip Flop: UGC NET CSE | July 2016 | Part 2 | Question: 10

In a positive edge triggered J-K flip flop , if J and K both are high then the output will be _____ on the rising edge of the clock :

- A. No change B. Set C. Reset D. Toggle

ugcnetcse-july2016-paper2 digital-logic flip-flop

Answer key 

8.9.7 Flip Flop: UGC NET CSE | July 2018 | Part 2 | Question: 92



In RS flip-flop, the output of the flip-flop at time $(t + 1)$ same as the output at time t , after the occurrence of a clock pulse if :

- A. $S = R = 1$ B. $S = 0, R = 1$ C. $S = 2, R = 0$ D. $S = R = 0$

ugcnetcse-july2018-paper2 digital-logic flip-flop

Answer key 

8.9.8 Flip Flop: UGC NET CSE | June 2009 | Part 2 | Question: 04



The characteristic equation of a JK flip-flop is :

- A. $Q_{n+1} = J \cdot Q_n + K \cdot Q_n$ B. $Q_{n+1} = J \cdot \bar{Q}_n + \bar{K} \cdot Q_n$
C. $Q_{n+1} = Q_n + J \cdot K$ D. $Q_{n+1} = (J + K)Q_n$

ugcnetcse-june2009-paper2 digital-logic flip-flop

Answer key 

8.9.9 Flip Flop: UGC NET CSE | September 2013 | Part 3 | Question: 34



The power dissipation of a flip-flop is 3 mW. The power dissipation of a digital system with 4 flip-flops is given by :

- A. 3^4 mW B. 4^3 mW C. $4/3$ mW D. 12 mW

ugcnetcse-sep2013-paper3 digital-logic flip-flop

Answer key 

8.9.10 Flip Flop: UGC NET CSE | September 2013 | Part 3 | Question: 35



An Astable multi vibrator using the 555 timer to generate a square wave of 5 KHz with 70% duty cycle will have :

- A. $R_A = 40.4K\Omega, R_B = 17.25K\Omega, C = 2000pF$
B. $R_A = 17.25K\Omega, R_B = 40.4K\Omega, C = 2000pF$
C. $R_A = 40.4K\Omega, R_B = 17.25K\Omega, C = 5000pF$
D. $R_A = 17.25K\Omega, R_B = 40.4K\Omega, C = 5000pF$

ugcnetcse-sep2013-paper3 digital-logic flip-flop

Answer key 

8.10 Floating Point Representation (2)

8.10.1 Floating Point Representation: UGC NET CSE | December 2014 | Part 2 | Question: 09



The range of representable normalized numbers in the floating point binary fractional representation in a 32-bit word with 1-bit sign, 8-bit excess 128 biased exponent and 23-bit mantissa is :

- A. 2^{-128} to $(1 - 2^{-23}) \times 2^{127}$ B. $(1 - 2^{-23}) \times 2^{-127}$ to 2^{128}
C. $(1 - 2^{-23}) \times 2^{-127}$ to 2^{23} D. 2^{-129} to $(1 - 2^{-23}) \times 2^{127}$

ugcnetcse-dec2014-paper2 digital-logic floating-point-representation

Answer key 

8.10.2 Floating Point Representation: UGC NET CSE | December 2018 | Part 2 | Question: 16



The decimal floating point number -40.1 represented using IEEE-754 32-bit representation and written in hexadecimal form is

- A. $0xC'2206666$ B. $0xC'2206000$ C. $0xC'2006666$ D. $0xC'2006000$

ugcnetcse-dec2018-paper2 digital-logic number-representation floating-point-representation

Answer key

8.11 Full Adder (1)

8.11.1 Full Adder: UGC NET CSE | Junet 2015 | Part 2 | Question: 7



Consider a full-adder with the following input values :

- a. $x = 1, y = 0$ and C_i (carry input)=0 b. $x = 0, y = 1$ and $C_i = 1$

Compute the value of S(sum) and C_o (carry output) for the above input values :

- A. $S = 1, C_o = 0$ and $S = 0, C_o = 1$ B. $S = 0, C_o = 0$ and $S = 1, C_o = 1$
C. $S = 1, C_o = 1$ and $S = 0, C_o = 0$ D. $S = 0, C_o = 1$ and $S = 1, C_o = 0$

ugcnetcse-june2015-paper2 digital-logic full-adder

Answer key

8.12 IEEE Representation (2)

8.12.1 IEEE Representation: UGC NET CSE | July 2016 | Part 2 | Question: 7



The IEEE-754 double precision format to represent floating point numbers, has a length of _____ bits.

- A. 16 B. 32 C. 48 D. 64

ugcnetcse-july2016-paper2 digital-logic ieee-representation

Answer key

8.12.2 IEEE Representation: UGC NET CSE | September 2013 | Part 2 | Question: 7



The IEEE single-precision and double-precision format to represent floating-point numbers, has a length of _____ and _____ respectively.

- A. 8 bits and 16 bits B. 16 bits and 32 bits
C. 32 bits and 64 bits D. 64 bits and 128 bits

ugcnetsep2013ii digital-logic ieee-representation

Answer key

8.13 K Map (6)

8.13.1 K Map: UGC NET CSE | August 2016 | Part 2 | Question: 8



The Karnaugh map for a Boolean function is given as

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	1	1	1	1
10	0	1	1	1

The simplified Boolean equation for the above Karnaugh Map is

- A. $AB + CD + A\bar{B} + AD$ B. $AB + AC + AD + BCD$
 C. $AB + AD + BC + ACD$ D. $AB + AC + BC + BCD$

ugcnetcse-aug2016-paper2 digital-logic k-map

Answer key 

8.13.2 K Map: UGC NET CSE | July 2018 | Part 2 | Question: 99



Simplify the following using K – map :

$$F(A, B, C, D) = \Sigma(0, 1, 2, 8, 9, 12, 13)$$

$$d(A, B, C, D) = \Sigma(10, 11, 14, 15)$$

d stands for don't care condition.

- A. $A + \bar{B}\bar{D} + BC$ B. $A + \bar{B}\bar{D} + \bar{B}\bar{C}$
 C. $\bar{A} + \bar{B}\bar{C}$ D. $\bar{A} + \bar{B}\bar{C} + \bar{B}\bar{D}$

ugcnetcse-july2018-paper2 digital-logic k-map

Answer key 

8.13.3 K Map: UGC NET CSE | June 2010 | Part 2 | Question: 10



The function represented by the k-map given below is

A \ BC	BC			
	00	01	11	10
0	1	0	0	1
1	1	0	0	1

- A. $A \cdot B$ B. $AB + BC + CA$
 C. $\overline{B \oplus C}$ D. $A \cdot B \cdot C$

ugcnetcse-june2010-paper2 digital-logic k-map

Answer key 

8.13.4 K Map: UGC NET CSE | June 2013 | Part 3 | Question: 45



The simplified function in product of sums of Boolean function $F(W, X, Y, Z) = \Sigma(0, 1, 2, 5, 8, 9, 10)$ is

- A. $(W' + X')(Y' + Z')(X' + Z)$ B. $(W' + X')(Y' + Z')(X' + Z')$
 C. $(W' + X')(Y' + Z)(X' + Z)$ D. $(W' + X')(Y + Z')(X' + Z)$

ugcnetcse-june2013-paper3 digital-logic k-map

Answer key 

8.13.5 K Map: UGC NET CSE | November 2017 | Part 2 | Question: 6



The Boolean function with the Karnaugh map

CD \ AB	00	01	11	10
	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	0

is

A. $(A + C) \cdot D + B$

B. $(A + B) \cdot C + D$

C. $(A + D) \cdot C + B$

D. $(A + C) \cdot B + D$

ugcnetcse-nov2017-paper2 digital-logic k-map

Answer key 

8.13.6 K Map: UGC NET CSE | October 2020 | Part 2 | Question: 42



Simplified expression/s for following Boolean function

$F(A, B, C, D) = \Sigma(0, 1, 2, 3, 6, 12, 13, 14, 15)$ is/are

i. $A'B' + AB + A'C'D'$

ii. $A'B' + AB + A'CD'$

iii. $A'B' + AB + BC'D'$

iv. $A'B' + AB + BCD'$

Choose the correct answer from the options given below:

A. (i) only

B. (ii) only

C. (i) and (ii) only

D. (ii) and (iv) only

ugcnetcse-oct2020-paper2 digital-logic k-map

Answer key 

8.14

Memory (1)

8.14.1 Memory: UGC NET CSE | December 2009 | Part 2 | Question: 08



The highest noise margin is offered by

A. BICMOS

B. TTL

C. ECL

D. CMOS

ugcnetcse-dec2009-paper2 digital-logic memory

8.15

Multiplexer (4)

8.15.1 Multiplexer: UGC NET CSE | December 2008 | Part 2 | Question: 15



If 4 input multiplexers drive a 4 input multiplexer, we get a:

A. 16 input MUX

B. 8 input MUX

C. 4 input MUX

D. 2 input MUX

ugcnetcse-dec2008-paper2 digital-logic multiplexer

Answer key 

8.15.2 Multiplexer: UGC NET CSE | June 2009 | Part 2 | Question: 05



In order to implement a n variable switching function, a MUX must have :

- A. 2^n inputs B. $2^n + 1$ inputs C. 2^{n-1} inputs D. $2^n - 1$ inputs

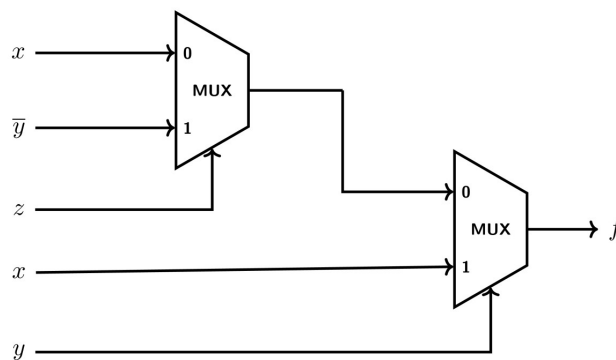
ugcnetcse-june2009-paper2 digital-logic multiplexer

Answer key 

8.15.3 Multiplexer: UGC NET CSE | June 2012 | Part 3 | Question: 53



Consider the below circuit and find the output function $f(x, y, z)$



- A. $x\bar{z} + xy + yz$ B. $x\bar{z} + xy + y\bar{z}$
C. $xz + xy + yz$ D. $xz + xy + yz$

ugcnetcse-june2012-paper3 digital-logic multiplexer

Answer key 

8.15.4 Multiplexer: UGC NET CSE | June 2014 | Part 3 | Question: 68



A terminal multiplexer has six 1200 bps terminals and ' n ' 300 bps terminals connected to it. If the outgoing line is 9600 bps, what is the value of n ?

- A. 4 B. 8 C. 16 D. 28

ugcnetjune2014iii digital-logic multiplexer

Answer key 

8.16

Number Representation (11)

8.16.1 Number Representation: UGC NET CSE | August 2016 | Part 2 | Question: 6



The octal number 326.4 is equivalent to

- A. $(214.2)_{10}$ and $(D6.8)_{16}$ B. $(212.5)_{10}$ and $(D6.8)_{16}$
C. $(214.5)_{10}$ and $(D6.8)_{16}$ D. $(214.5)_{10}$ and $(D6.4)_{16}$

ugcnetcse-aug2016-paper2 digital-logic number-representation

Answer key 

8.16.2 Number Representation: UGC NET CSE | August 2016 | Part 2 | Question: 7



Which of the following is the most efficient to perform arithmetic operations on the numbers ?

- A. Sign-magnitude
C. 2's complement

- B. 1's complement
D. 9's complement

ugcnetcse-aug2016-paper2 digital-logic number-representation

Answer key 

8.16.3 Number Representation: UGC NET CSE | December 2009 | Part 2 | Question: 09



The answer of the operation $(10111)_2 * (1110)_2$ in hex equivalence is

- (A) 150
(B) 241
(C) 142
(D) 101011110

ugcnetcse-dec2009-paper2 digital-logic number-representation

Answer key 

8.16.4 Number Representation: UGC NET CSE | December 2009 | Part 2 | Question: 10



How many 1's are present in the binary representation of $3 \times 512 + 7 \times 64 + 5 \times 8 + 3$

- (A) 8
(B) 9
(C) 10
(D) 11

ugcnetcse-dec2009-paper2 digital-logic number-representation

Answer key 

8.16.5 Number Representation: UGC NET CSE | December 2010 | Part 2 | Question: 6



The decimal number equivalent of $(4057.06)_8$ is

- A. 2095.75 B. 2095.075 C. 2095.937 D. 2095.0937

ugcnetcse-dec2010-paper2 digital-logic number-representation

Answer key 

8.16.6 Number Representation: UGC NET CSE | December 2010 | Part 2 | Question: 9



12-bit 2's complement of -73.75 is

- A. 01001001.1100 B. 11001001.1100
C. 10110110.0100 D. 10110110.1100

ugcnetcse-dec2010-paper2 digital-logic number-representation

Answer key 

8.16.7 Number Representation: UGC NET CSE | July 2018 | Part 2 | Question: 95



The hexadecimal equivalent of the binary integer number 110101101 is

- A. D24 B. 1BD C. 1AE D. 1AD

ugcnetcse-july2018-paper2 digital-logic number-representation easy

Answer key 

8.16.8 Number Representation: UGC NET CSE | July 2018 | Part 2 | Question: 96

Perform the following operation for the binary equivalent of the decimal numbers $(-14)_{10} + (-15)_{10}$. The solution in 8 bit representation is

- A. 11100011 B. 00011101 C. 10011101 D. 11110011

ugcnetcse-july2018-paper2 digital-logic number-representation

Answer key

8.16.9 Number Representation: UGC NET CSE | June 2005 | Part 2 | Question: 9

Which of the following binary number is the same as its $2'$ s complement :

- A. 1010 B. 0101 C. 1000 D. 1001

ugcnetcse-june2005-paper2 digital-logic number-representation

Answer key

8.16.10 Number Representation: UGC NET CSE | June 2012 | Part 2 | Question: 35

If an integer needs two bytes of storage, then the maximum value of signed integer is

- A. $2^{16} - 1$ B. $2^{15} - 1$ C. 2^{16} D. 2^{15}

digital-logic number-representation ugcnetcse-june2012-paper2

Answer key

8.16.11 Number Representation: UGC NET CSE | June 2014 | Part 2 | Question: 18

Which of the following statement(s) is (are) not correct ?

- The $2'$ s complement of 0 is 0.
- In $2'$ s complement, the left most bit cannot be used to express a quantity.
- For an n -bit word ($2'$ s complement) which includes the sign bit, there are 2^{n-1} positive integers, 2^{n+1} negative integers and one 0 for a total of 2^n unique states.
- In $2'$ s complement the significant information is contained in the 1's of positive numbers and 0's of the negative numbers.

- A. *i & iv* B. *i & ii* C. *iii* D. *iv*

ugcnetcse-june2014-paper2 digital-logic number-representation

Answer key

8.17**Number System (10)****8.17.1 Number System: UGC NET CSE | December 2005 | Part 2 | Question: 6**

Which of the following is divisible by 4?

- A. 100101100 B. 1110001110001 C. 11110011 D. 10101010101010

ugcnetcse-dec2005-paper2 digital-logic number-system

Answer key

8.17.2 Number System: UGC NET CSE | December 2006 | Part 2 | Question: 6

The hexadecimal equivalent of $(10111)_2 \times (1110)_2$ is :

- A. 150 B. 241 C. 142 D. 101011110

ugcnetcse-dec2006-paper2 digital-logic number-system

Answer key 

8.17.3 Number System: UGC NET CSE | December 2013 | Part 2 | Question: 32



Given that $(292)_{10} = (1204)_x$ in some number system x . The base x of that number system is

- A. 2 B. 8 C. 10 D. None of the above

ugcnetcse-dec2013-paper2 digital-logic number-system

Answer key 

8.17.4 Number System: UGC NET CSE | January 2017 | Part 2 | Question: 10



The hexadecimal equivalent of the octal number 2357 is :

- A. 2EE B. 2FF C. 4EF D. 4FE

ugcnetjan2017ii digital-logic number-system

Answer key 

8.17.5 Number System: UGC NET CSE | January 2017 | Part 2 | Question: 15



If X is a binary number which is power of 2, then the value of $X \& (X - 1)$ is :

- A. 11...11 B. 00...00 C. 100...0 D. 000...1

ugcnetjan2017ii digital-logic number-system

Answer key 

8.17.6 Number System: UGC NET CSE | January 2017 | Part 2 | Question: 9



Convert the octal number 0.4051 into its equivalent decimal number.

- A. 0.5100098 B. 0.2096 C. 0.52 D. 0.4192

ugcnetjan2017ii digital-logic number-system

Answer key 

8.17.7 Number System: UGC NET CSE | June 2009 | Part 2 | Question: 13



The octal equivalent of hexadecimal $(A.B)_{16}$ is :

- A. 47.21 B. 12.74 C. 12.71 D. 17.21

ugcnetcse-june2009-paper2 digital-logic number-system number-representation

Answer key 

8.17.8 Number System: UGC NET CSE | June 2019 | Part 2 | Question: 18



Consider the equation $(146)_b + (313)_{b-2} = (246)_8$. Which of the following is the value of b ?

- A. 8 B. 7 C. 10 D. 16

ugcnetcse-june2019-paper2 number-system

Answer key 

8.17.9 Number System: UGC NET CSE | Junet 2015 | Part 3 | Question: 4



The equivalent hexadecimal notation for octal number 2550276 is

- A. FADED B. AEOBE C. ADOBE D. ACABE

ugcnetcse-june2015-paper3 digital-logic number-system

Answer key 

8.17.10 Number System: UGC NET CSE | November 2017 | Part 2 | Question: 7



The octal equivalent of the binary number 1011101011 is

- A. 7353 B. 1353 C. 5651 D. 5657

ugcnetcse-nov2017-paper2 digital-logic number-system number-representation

Answer key 

8.18 Number Theory (1)

8.18.1 Number Theory: UGC NET CSE | June 2010 | Part 2 | Question: 9



What is decimal equivalent of BCD 11011.1100?

- A. 22.0 B. 22.2 C. 20.2 D. 21.2

ugcnetcse-june2010-paper2 digital-logic number-theory

Answer key 

8.19 Priority Encoder (1)

8.19.1 Priority Encoder: UGC NET CSE | June 2019 | Part 2 | Question: 17



The parallel bus arbitration technique uses an external priority encoder and a decoder. Suppose, a parallel arbiter has 5 bus arbiters. What will be the size of priority encoder and decoder respectively?

- A. $4 \times 2, 2 \times 4$ B. $2 \times 4, 4 \times 2$
C. $3 \times 8, 8 \times 3$ D. $8 \times 3, 3 \times 8$

ugcnetcse-june2019-paper2 priority-encoder

Answer key 

8.20 ROM (1)

8.20.1 ROM: UGC NET CSE | December 2014 | Part 2 | Question: 10



The size of the *ROM* required to build an 8-bit adder / subtractor with mode control, carry input, carry output and two's complement overflow output is given as

- A. $2^{16} \times 8$ B. $2^{18} \times 10$
C. $2^{16} \times 10$ D. $2^{18} \times 8$

ugcnetcse-dec2014-paper2 digital-logic rom

Answer key 

8.21 Ripple Counter Operation (3)

8.21.1 Ripple Counter Operation: UGC NET CSE | August 2016 | Part 3 | Question: 1



A ripple counter is a (n):

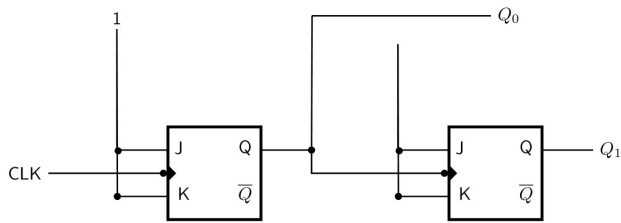
- A. Synchronous Counter B. Asynchronous counter
C. Parallel counter D. None of the above

ugcnetcse-aug2016-paper3 digital-logic ripple-counter-operation

8.21.2 Ripple Counter Operation: UGC NET CSE | July 2018 | Part 2 | Question: 94



What does the following logic diagram represent?



- A. Synchronous Counter
C. Combinational Circuit

- B. Ripple Counter
D. Mod 2 Counter

ugcnetcse-july2018-paper2 digital-logic ripple-counter-operation

Answer key

8.21.3 Ripple Counter Operation: UGC NET CSE | September 2013 | Part 3 | Question: 36



A binary ripple counter is required to count up to 16383. How many flip-flops are required?

- A. 16382 B. 8191 C. 512 D. 14

ugcnetcse-sep2013-paper3 digital-logic ripple-counter-operation

Answer key

8.22 Sequential Circuit (6)

8.22.1 Sequential Circuit: UGC NET CSE | July 2016 | Part 3 | Question: 1



Which of the following is a sequential circuit?

- A. Multiplexer B. Decoder C. Full adder D. Counter

ugcnetcse-july2016-paper3 digital-logic sequential-circuit

Answer key

8.22.2 Sequential Circuit: UGC NET CSE | June 2009 | Part 2 | Question: 14



A reduced state table has 18 rows. The minimum number of Flip-flops needed to implement the sequential machine is :

- A. 18 B. 9 C. 5 D. 4

ugcnetcse-june2009-paper2 digital-logic sequential-circuit

Answer key

8.22.3 Sequential Circuit: UGC NET CSE | June 2010 | Part 2 | Question: 7



Advantage of synchronous sequential circuits over asynchronous ones is

- A. Faster operation B. Ease of avoiding problems due to hazard
C. Lower hardware requirement D. Better noise immunity

ugcnetcse-june2010-paper2 digital-logic sequential-circuit

Answer key

8.22.4 Sequential Circuit: UGC NET CSE | June 2014 | Part 3 | Question: 42



The output of a sequential circuit depends on

- A. Present input only B. Past input only
C. Both present and past input D. Past output only

ugcnetjune2014iii digital-logic sequential-circuit

Answer key

8.22.5 Sequential Circuit: UGC NET CSE | June 2014 | Part 3 | Question: 44

Which of the following flip-flops is free from race condition ?

- A. T flip-flop
 B. SR flip-flop
 C. Master-slave JK flip-flop
 D. None of the above

ugcnetjune2014iii digital-logic sequential-circuit flip-flop

Answer key

8.22.6 Sequential Circuit: UGC NET CSE | Junet 2015 | Part 3 | Question: 6

The number of flip-flops required to design a modulo -272 counter is

- A. 8
 B. 9
 C. 27
 D. 11

ugcnetcse-june2015-paper3 digital-logic sequential-circuit flip-flop

Answer key

8.23 Sum of Product (1)**8.23.1 Sum of Product: UGC NET CSE | December 2013 | Part 2 | Question: 33**

The sum of products expansion for the function $F(x, y, z) = (x + y)\bar{z}$ is given as

- A. $\bar{x}\bar{y}z + xy\bar{z} + \bar{x}y\bar{z}$
 B. $xyz + xy\bar{z} + \bar{x}y\bar{z}$
 C. $\bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + xy\bar{z}$
 D. $xy\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z}$

ugcnetcse-dec2013-paper2 digital-logic sum-of-product

Answer key

8.24 Unix (1)**8.24.1 Unix: UGC NET CSE | December 2015 | Part 3 | Question: 40**

In Unix operating system, special files are used to

- A. buffer data received in its input from where a process reads
 B. provide a mechanism to map physical device to file names
 C. store list of file names plus pointers associated with i-nodes
 D. store information entered by a user application program or utility program

ugcnetcse-dec2015-paper3 unix

Answer key

Answer Keys

8.0.1	C	8.0.2	A	8.0.3	A	8.0.4	C	8.0.5	A
8.0.6	C	8.0.7	C	8.0.8	Q-Q	8.0.9	B	8.0.10	Q-Q
8.0.11	D	8.0.12	Q-Q	8.0.13	Q-Q	8.0.14	Q-Q	8.0.15	Q-Q
8.0.16	Q-Q	8.0.17	C	8.1.1	Q-Q	8.1.2	Q-Q	8.1.3	D
8.1.4	D	8.2.1	C	8.2.2	Q-Q	8.2.3	A	8.2.4	Q-Q
8.2.5	C	8.2.6	C	8.2.7	D	8.2.8	D	8.2.9	D
8.2.10	D	8.3.1	A;D	8.3.2	C	8.3.3	C	8.3.4	Q-Q

8.3.5	B
8.5.2	Q-Q
8.5.7	B;D
8.6.2	B
8.9.3	Q-Q
8.9.8	Q-Q
8.11.1	A
8.13.3	Q-Q
8.15.1	Q-Q
8.16.2	Q-Q
8.16.7	Q-Q
8.17.1	Q-Q
8.17.6	A
8.18.1	Q-Q
8.21.3	D
8.22.5	C

8.4.1	Q-Q
8.5.3	Q-Q
8.5.8	B
8.7.1	B
8.9.4	Q-Q
8.9.9	D
8.12.1	D
8.13.4	A
8.15.2	Q-Q
8.16.3	C
8.16.8	Q-Q
8.17.2	Q-Q
8.17.7	Q-Q
8.19.1	D
8.22.1	C
8.22.6	B

8.4.2	Q-Q
8.5.4	Q-Q
8.5.9	Q-Q
8.8.1	A OR C
8.9.5	D
8.9.10	X
8.12.2	C
8.13.5	A
8.15.3	A
8.16.4	Q-Q
8.16.9	Q-Q
8.17.3	D
8.17.8	B
8.20.1	B
8.22.2	Q-Q
8.23.1	D

8.4.3	Q-Q
8.5.5	A
8.5.10	C
8.9.1	Q-Q
8.9.6	D
8.10.1	D
8.13.1	B
8.13.6	D
8.15.4	B
8.16.5	Q-Q
8.16.10	B
8.17.4	C
8.17.9	C
8.21.1	Q-Q
8.22.3	Q-Q
8.24.1	D

8.5.1	Q-Q
8.5.6	A
8.6.1	Q-Q
8.9.2	Q-Q
8.9.7	Q-Q
8.10.2	Q-Q
8.13.2	Q-Q
8.14.1	Q-Q
8.16.1	Q-Q
8.16.6	Q-Q
8.16.11	C
8.17.5	B
8.17.10	B
8.21.2	Q-Q
8.22.4	C